Construction

Step 1: With centre at O, draw a circle of radius 3 cm.

Step 2: Draw a line OP of length 8 cm.

Step 3: Draw a perpendicular bisector of OP, which cuts OP at M.

Step 4: With M as centre and MO as radius, draw a circle which cuts previous circle at A and B.

Step5: Join AP and BP. AP and BP are the required tangents. Thus length of the tangents are $PA = PB = 7.4$ cm.

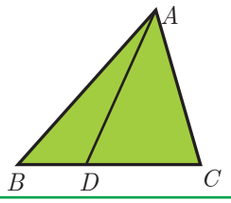
Verification : In the right angle triangle OAP, $PA^2 = OP^2 - OA^2 = 8^2 - 3^2 = 64 - 9 = 55$

$PA = \sqrt{55} = 7.4$ cm (approximately) .

4.6 Concurrency Theorems

Definition

A cevian is a line segment that extends from one vertex of a triangle to the opposite side. In the diagram, AD is a cevian, from A.



Special cevians

(i) A median is a cevian that divides the opposite side into two congruent(equal) lengths.

(ii) An altitude is a cevian that is perpendicular to the opposite side.

(iii) An angle bisector is a cevian that bisects the corresponding angle.

DO YOU KNOW?

The term cevian comes from the name of Italian engineer Giovanni Ceva, who proved a well known theorem about cevians.

Ceva's Theorem (without proof)

Statement

Let ABC be a triangle and let D, E, F be points on lines BC, CA, AB respectively. Then the cevians AD, BE, CF are concurrent if and only if $\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$ where the lengths are directed. This also works for the reciprocal of each of the ratios as the reciprocal of 1 is 1.

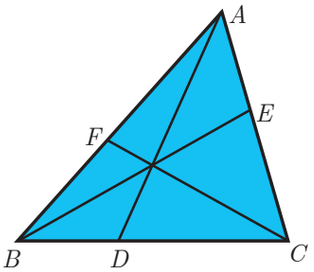


Fig. 4.70

DO YOU KNOW?

Giovanni Ceva (Dec 7, 1647 – June 15, 1734)

In 1686, Ceva was designated as the professor of Mathematics, University of Mantua and worked there for the rest of the life. In 1678, he published an important theorem on synthetic geometry for a triangle called Ceva's theorem.

Ceva also rediscovered and published in the Journal Opuscula mathematica and Geometria motus in 1692. He applied these ideas in mechanics and hydraulics.

Note

The cevians do not necessarily lie within the triangle, although they do in the diagram.

10th_Maths_Chapter 4_English.indd 195 12/27/2024 3:11:36 PM